

DistillToMCU

behavior distillation for LLM-free MCU autonomy

Cloud LLM

DeepSeek / Qwen
tool-call decisions
per sensor event

COMIC distillation

online quantiles
conformal calibration
MDL consolidation

ESP32-S3

local rule matching
1.48 ms (p50)
no LLM at runtime

temp in [22.5, 28.0]
conf 0.92 · 2 B/rule

● 85.4% held-out autonomy

● 50.0% precision on UCI replay

● teacher shift 1.7–3.7